

RA-SCOPE: An Auditable Zero-Parameter-Fusion Ensemble for Ordinal Difficulty Assessment of Second-Language Chinese Graded Readers

Tingrui You¹, Na Zhang², Ling Xiong², and Jimei Li^{2,*}

Beijing Language and Culture University

¹Hainan International College ²School of Information Science

Beijing, China

Emails: yyt2006@163.com; zn17333099816@163.com; lingdonghua99@163.com; Ljm@blcu.edu.cn

*Corresponding author

Abstract—Assessing the difficulty level of second-language (L2) reading materials is a recurring educational-measurement task that currently depends on costly expert panels and produces results that are fragile across publishing platforms. We present RA-SCOPE, an auditable inference layer for ordinal, book-level difficulty assessment of L2 Chinese graded picturebooks. Four heterogeneous monotone experts (length, character n-gram, Standard-derived measurements, and a fine-tuned Chinese MacBERT) are combined by an unweighted probability pool that introduces no trainable fusion parameters. A label-free source-distance support rank licenses a closed-form Euclidean projection of the pooled logits onto a retained-margin ordinal set. The projection is applied only when the two complementary anchors agree and at least one of them is backed by source evidence. Every item therefore carries an explicit state (CONSENSUS, CONFLICT, or NO-SUPPORT) that a human review workflow can act on. On 120 independently triple-annotated and adjudicated Chinese picturebooks, RA-SCOPE reaches .823 quadratic weighted kappa and .417 MAE, improving on fine-tuned MacBERT by .060 and on a Qwen2.5-3B rubric baseline by .285 with paired 95% intervals that exclude zero, and it halves the rate of errors exceeding one level. It retains .786 QWK and .393 MAE with no errors above one level on a separately adjudicated 28-book collection. A post-hoc stacked logistic fusion reaches .844 QWK and .277 MAE where target labels exist, trading the zero-parameter property and the preservation guarantee for lower absolute error. Every activated consensus grade is preserved by construction, and every decision can be re-executed by hand.

Index Terms—educational measurement, ordinal regression, ensemble inference, readability assessment, auditable machine learning, human-in-the-loop review

I. INTRODUCTION

Placing a graded reader at the right level is a routine but consequential educational-measurement decision. Publishers, reading programs and curriculum teams must decide whether a book belongs at level 1 or level 4, and the decision is made under a published standard (the Chinese Graded Readers Standards for International Chinese Language Education [1]) rather than by preference.

Today the decision is made by expert panels: the collections used in this study were graded by three annotators with International Chinese Education training who each contributed approximately 20 hours, followed by adjudication by a fourth.

The cost of that process does not scale with catalog growth, and its output is fragile across sources: books that look comparable in length are graded differently by different publishers because layout, illustration and annotation conventions differ sharply.

Automatic alternatives are available, but average fit is not what a review workflow needs. A fine-tuned Chinese encoder reaches .763 quadratic weighted kappa (QWK) on our main collection, and a deterministic Qwen2.5-3B rubric prompt reaches only .538 while showing strong sensitivity to surface length. More importantly, neither returns a decision that a reviewer can inspect: when such a model changes a grade, nothing in its output says whether the change is supported by evidence resembling the source material or whether it is an extrapolation.

Ensembles of ordinal experts make the situation sharper rather than simpler. Pooling heterogeneous ordinal predictors is known to help, and projection methods repair order violations, but the literature does not specify which evidence-backed ordinal decisions a heterogeneous ensemble is permitted to overturn. That question, rather than the pooling arithmetic, is what determines whether an automated grade can be adopted in practice.

We address it with RA-SCOPE (Reliability-Anchored Support-Backed Consensus Ordinal Projection Ensemble). Four experts first form a monotone probability pool. Length and character n-gram experts act as complementary anchors, and source-only nearest-neighbour distances become finite-sample support ranks. A projection is licensed only when both anchors predict the same grade and at least one of them has sufficient support. It is the unique Euclidean projection of the pooled logits onto a retained-margin ordinal set, with a closed-form solution and no fitted fusion parameters. Fig. 1 summarizes the design.

Our contributions are:

- a lineage-disjoint, multi-source evaluation protocol with independent triple annotation, separate adjudication of every disagreement, a held-out robustness collection,

source-only out-of-fold results, and paired lineage-bootstrap intervals;

- a support-backed full-consensus activation rule that never promotes a single supported anchor to a multi-anchor consensus;
- a closed-form constrained projection with proofs of monotonicity, uniqueness, and activated-grade preservation; and
- analyses of support states, hyperparameter sensitivity, length residuals, counter-length pairs, Standard-derived construct alignment, observable-modality routing, and a learned-fusion comparison that quantifies the price of auditability.

RA-SCOPE is an inference layer rather than a replacement model. It consumes the cumulative probability chains that the four experts already emit, introduces no trainable fusion parameters, and returns for each book an ordinal distribution together with one of three states. That output is what allows the system to be placed in front of, or behind, an existing human review queue: Section VII sets out the resulting triage behavior and its measured cost.

II. RELATED WORK

A. Readability and Ordinal Modeling

Automatic readability assessment has moved from lexical, syntactic, discourse and psycholinguistic measurements [2], [3] to hybrid models that combine pretrained encoders with handcrafted features [4]–[7]. Ordinal objectives exploit label order [8], [9], and cumulative rank models enforce ordered thresholds [10]. Chinese work includes Standard-derived feature quantification [11], adaptive pretraining with feature fusion [12], and book-level long-text modeling [13]. We focus on inference-time composition and decision guarantees rather than on another encoder objective.

B. Ordinal Ensembles and Projection

Projection-based ordinal ensembles decompose learning into ordered classification hypotheses [14], and isotonic projection repairs threshold order [15]. In contrast, RA-SCOPE projects an already-monotone heterogeneous pool at inference time onto an item-specific feasible set derived from anchor agreement and retained margins. The constraint therefore governs decision compatibility, not merely rank consistency.

C. Support and Selective Prediction

Distance in a pretrained representation space is a strong non-parametric signal for distributional support [16], and conformal ranks convert arbitrary nonconformity scores into marginally valid p -values under exchangeability [17]. Calibration and selective prediction address uncertainty or abstention [18]–[21]; RA-SCOPE instead always returns an ordinal distribution while exposing CONSENSUS, CONFLICT and NO-SUPPORT states.

TABLE I
LINEAGE-DISJOINT EVIDENCE COLLECTIONS

Collection	n	Role
Source development	96	Expert fitting; support calibration
Main	120	Adjudicated evaluation
Robustness	28	Separate adjudicated evaluation
Upper-band	51	Unscored support audit

III. TASK AND EVIDENCE

A. Book-Level Task

For book i the primary target is $y_i \in \{1, \dots, 6\}$. Ten secondary criteria operationalise the Standard: character, vocabulary, grammar/sentence, discourse, semantic-content, character/plot, genre/theme, illustration independence, pinyin/annotation reduction, and layout density. The first seven describe linguistic or content demand; the final three describe access and presentation. Proficiency inventories follow GF 0025–2021 [22]. Each annotation records a best level, a plausible interval, and evidence pages where needed.

B. Books, Independence, and Adjudication

Books come from StoryWeaver [23], the Global African Storybook Project [24], and the Global Digital Library [25]. Original-work lineages were audited before analysis; no book or lineage crosses the source-development and evaluation collections (Table I). Annotators P01–P03 independently annotated randomized spreadsheets; P04 then adjudicated disagreements and checked the original sources, and is excluded from inter-rater reliability. All four had International Chinese Education training and each contributed approximately 20 hours. On the 120-book main collection, ordinal Krippendorff’s α [26] is .677 (book-bootstrap 95% CI [.594, .740]) for global grades and .586 [.524, .638] over 1,200 criterion units. Adjudicated global counts for L1–L6 are 11/35/40/30/4/0. The 28-book robustness collection is 7/15/4/2/0/0, and its adjudicated global labels equal the three-rater medians for every book.

IV. SUPPORT-BACKED CONSENSUS PROJECTION

Let threshold $k \in \{2, \dots, 6\}$ denote $P(Y_i \geq k)$. Expert e emits a cumulative chain p_{ik}^e , projected with PAVA so that $p_{i2}^e \geq \dots \geq p_{i6}^e$. Its point grade is

$$g_i^e = 1 + \sum_{k=2}^6 \mathbb{I}[p_{ik}^e \geq .5]. \quad (1)$$

A. Complementary Monotone Experts

The four experts are: (1) a log-character-count ordinal model; (2) character TF-IDF with 2–5-grams [27]; (3) an ordinal model over 16 Standard-derived measurements, including lexical inventories, sentence and page density, dialogue, and diversity; and (4) page-pooled Chinese MacBERT [28] with a cumulative ordinal head. Five-fold lineage-grouped, source-only cross-validation selects hyperparameters. MacBERT uses

three complete seeds, three epochs, and four sampled pages per book and epoch. The uniform pool is

$$\bar{p}_{ik} = \frac{1}{4} \sum_{e=1}^4 p_{ik}^e, \quad (2)$$

which introduces no trainable fusion parameters.

B. Label-Free Support Rank

Length (L) and character TF-IDF (C) are the anchors, $\mathcal{A} = \{L, C\}$. For each anchor, lineage-held-out source episodes yield nearest-neighbour representation distances d_j^e , $j = 1, \dots, M$. For a new item,

$$u_i^e = \frac{\sum_{j=1}^M \mathbb{1}[d_j^e \leq d_i^e]}{M+1}, \quad s_i^e = 1 - u_i^e. \quad (3)$$

A large u indicates a source-distance tail; s is its support rank. We mark anchor e as support-eligible when $u_i^e \leq .70$ (equivalently $s_i^e \geq .30$). With continuous distances, s is the usual conformal rank p -value; under exchangeability it is marginally super-uniform [17]. This conditional result licenses a support diagnostic, not a correctness probability. The projection activates exactly when

$$a_i = \mathbb{1}\left[g_i^L = g_i^C \wedge \max_{e \in \mathcal{A}} s_i^e \geq .30\right]. \quad (4)$$

Thus agreement always concerns both anchors; one supported anchor cannot create consensus. If supported anchors exist but $g_i^L \neq g_i^C$, the state is CONFLICT; if neither qualifies, it is NO-SUPPORT.

C. A Closed-Form Projection

Let $z_{ik}^e = \text{logit}(p_{ik}^e)$ and $\bar{z}_{ik} = \text{logit}(\bar{p}_{ik})$, with clipping only at the logit boundary. For an active item, write the shared grade as g_i^* and retain fraction $\lambda = .5$. Define the convex feasible set

$$\mathcal{C}_i = \left\{ z : z_2 \geq \dots \geq z_6, z_k \geq \lambda \max_{e \in \mathcal{A}} z_{ik}^e (k \leq g_i^*), z_k \leq \lambda \min_{e \in \mathcal{A}} z_{ik}^e (k > g_i^*) \right\} \quad (5)$$

RA-SCOPE returns

$$z_i^* = \arg \min_{z \in \mathcal{C}_i} \frac{1}{2} \|z - \bar{z}_i\|_2^2, \quad (6)$$

and otherwise returns $\bar{z}_i$. The solution is coordinate-wise clipping:

$$z_{ik}^* = \begin{cases} \max(\bar{z}_{ik}, \lambda \max_e z_{ik}^e), & k \leq g_i^*, \\ \min(\bar{z}_{ik}, \lambda \min_e z_{ik}^e), & k > g_i^*. \end{cases} \quad (7)$$

Proposition 1 (valid ordinal output). *The pool in (2) and the projected probabilities $\sigma(z_i^*)$ are non-increasing cumulative chains.*

Proof. Convex combinations preserve every adjacent inequality. In an active case, the pooled chain and the maxima/minima of anchor chains are non-increasing. Clipping two non-increasing sequences is non-increasing within the shared positive prefix and negative suffix; their sign boundary cannot invert. ■

Proposition 2 (unique consensus preservation). *For $0 < \lambda \leq 1$, (7) is the unique Euclidean projection in (6), and its point grade is g_i^* .*

Proof. The objective is strictly convex and $\mathcal{C}_i$ is non-empty, closed and convex [29]. Coordinate clipping is the Euclidean projection onto the retained-margin box; Proposition 1 places that solution inside its monotone intersection, so it also solves (6). Every prefix bound is non-negative and every suffix bound is negative, preserving all threshold signs and therefore g_i^* . ■

V. EXPERIMENTAL PROTOCOL

Comparators. We report each expert and the unprojected uniform pool. A contemporary prompt baseline uses Qwen2.5-3B-Instruct [30]: a fixed Chinese six-level rubric, the full extracted book text, and restricted next-token scoring over the six level tokens. It has no demonstrations, generation sampling, or target-label selection. Projection ablations require both anchors to be supported, activate on an agreeing qualified subset, or remove the support condition while retaining full agreement.

Metrics and uncertainty. The primary metric is quadratic weighted kappa (QWK) [31]. We also report MAE, exact accuracy, within-one accuracy, and errors above one level. Probability diagnostics comprise threshold-averaged ordinal Brier score, class negative log-likelihood (NLL), 10-bin expected calibration error (ECE), and MAE risk-coverage area (RC-AUC); lower is better. All method differences use 10,000 paired percentile-bootstrap samples over original-work lineages [32]. Support cutoffs $u \in \{.50, .60, .70, .80, .90, .95\}$ and margin relaxations in $\{.25, .50, .75\}$ test sensitivity.

Reproducibility. The PyTorch module consumes raw cumulative probabilities and exposes every support state. Tests cover monotonicity, full-consensus activation, single-supported-anchor disagreement, unsupported agreement, preservation, invalid inputs, and zero trainable parameters. The artifact records source folds, per-book predictions, bootstrap outputs, exact model identifiers, prompt hashes, and SHA-256 inventories.

VI. RESULTS

A. Global Grading

RA-SCOPE reaches .823 QWK and .417 MAE on the main collection (Table II). Two properties matter more for deployment than the QWK delta. First, the rate of errors exceeding one level falls from .017 for the uniform pool to .008, halving a failure mode that is far costlier than an off-by-one grade. Second, the gains over the strongest individual comparator are substantial and their intervals are far from zero: +.060 [.008, .118] QWK and −.142 [−.250, −.033] MAE against fine-tuned MacBERT, and +.285 [.173, .403] QWK and −.408 [−.600, −.217] MAE against the Qwen2.5-3B rubric baseline. Relative to uniform pooling, itself a strong heterogeneous baseline, the direct QWK gain is +.032 [.005, .066] with an MAE change of −.042 [−.092, .008].

Source-only out-of-fold predictions obtain .858 QWK and .406 MAE with one error above one level among 96 books. On the separate robustness collection RA-SCOPE obtains .786

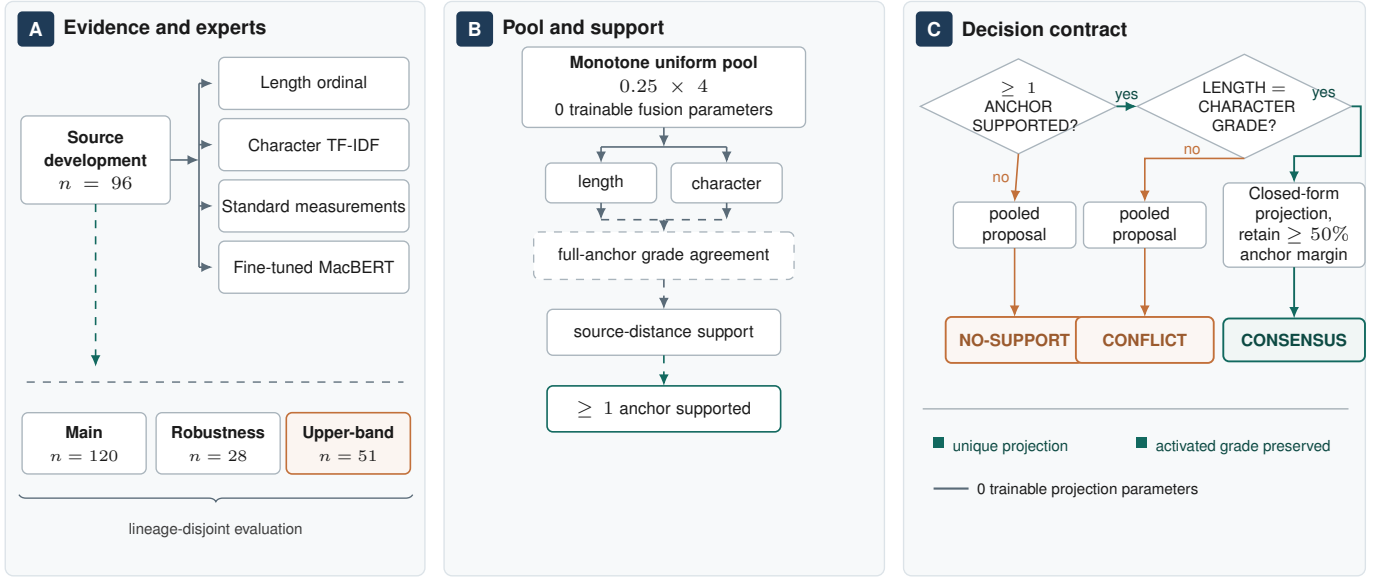


Fig. 1. RA-SCOPE dataflow. The source-development collection trains the experts and calibrates the support rank. Four monotone experts form a parameter-free uniform pool. The decision contract first checks whether at least one anchor is source-supported, then whether the length and character grades agree; grade agreement always involves both anchors, and support from a single anchor can license an existing agreement but never create one. Supported agreement activates projection and returns CONSENSUS; supported disagreement retains the pool and returns CONFLICT; absence of support retains the pool and returns NO-SUPPORT.

TABLE II
GLOBAL GRADING. THE SUPPORT-DISABLED ABLATION APPLIES PROJECTION ON FULL ANCHOR AGREEMENT WITHOUT THE SUPPORT CONDITION; IT GIVES IDENTICAL POINT PREDICTIONS TO RA-SCOPE ON BOTH COLLECTIONS.

Method	Main ($n=120$)				Rob. ($n=28$)	
	QWK	MAE	Exact	> 1	QWK	MAE
Character TF-IDF	.577	.708	.425	.125	.487	.786
Length ordinal	.767	.542	.492	.025	.772	.429
Standard measurements	.622	.633	.467	.083	.620	.536
MacBERT (3 seeds)	.763	.558	.467	.025	.672	.571
Qwen2.5-3B rubric	.538	.825	.425	.183	.603	.714
Uniform probability pool	.790	.458	.558	.017	.786	.393
RA-SCOPE	.823	.417	.592	.008	.786	.393
Support disabled	.823	.417	.592	.008	.786	.393

QWK, .393 MAE, 60.7% exact accuracy and 100% within one level; its gains over MacBERT and Qwen are +.114 [.030, .227] and +.183 [.008, .313] QWK respectively. The interval against length includes zero, as expected for 28 books.

B. What the Support Contract Guarantees

The support contract is a decision-audit mechanism rather than an accuracy lever. On these collections the three unsupported agreements do not cross a decision threshold, so disabling the support condition yields identical point grades (Table II, final row); requiring both anchors to be supported gives .815/.425 and allowing a singleton qualified subset to activate gives .799/.450. The contract’s value appears instead in the state distribution it exposes. The main collection contains 38 support-backed full consensus, 46 supported conflicts, and 36 books without a supported anchor, with exact

TABLE III
SUPPORT STATES ON THE MAIN COLLECTION. EVERY ROUTE RETURNS A GRADE.

State	n	Exact	Route
CONSENSUS	38	.711	Projection applied; grade preserved
CONFLICT	46	.565	Pooled proposal retained
NO-SUPPORT	36	.500	Pooled proposal retained

accuracy .711, .565 and .500 respectively. The state is therefore informative even though every route returns a prediction, and it is exact rather than fitted: the projection preserves 38/38 activated grades on the main collection, 32/32 on source out-of-fold predictions, and 7/7 on the robustness collection. Across support cutoffs, main QWK stays in [.817, .825], and every tested retained margin produces identical point grades.

C. Learned Fusion Baseline

RA-SCOPE is deliberately zero-parameter, so the natural objection is that a fitted fusion over the same experts would do better. As a post-hoc analysis we test this directly. Using the four experts’ continuous ordinal scores from the frozen artifact (four features per book: length, character TF-IDF, Standard-derived measurements, and fine-tuned MacBERT), we fit a multinomial logistic regression on the 120-book collection. Folds are grouped by original-work lineage, matching the main protocol; every lineage holds a single evaluated book on this collection, so grouped and stratified folds coincide. We repeat the cross-validation over ten seeds at $k \in \{4, 5, 6, 10\}$ and report the mean over the ten seeds at $k=5$, which Table IV gives. Learned fusion is competitive on QWK and clearly better on

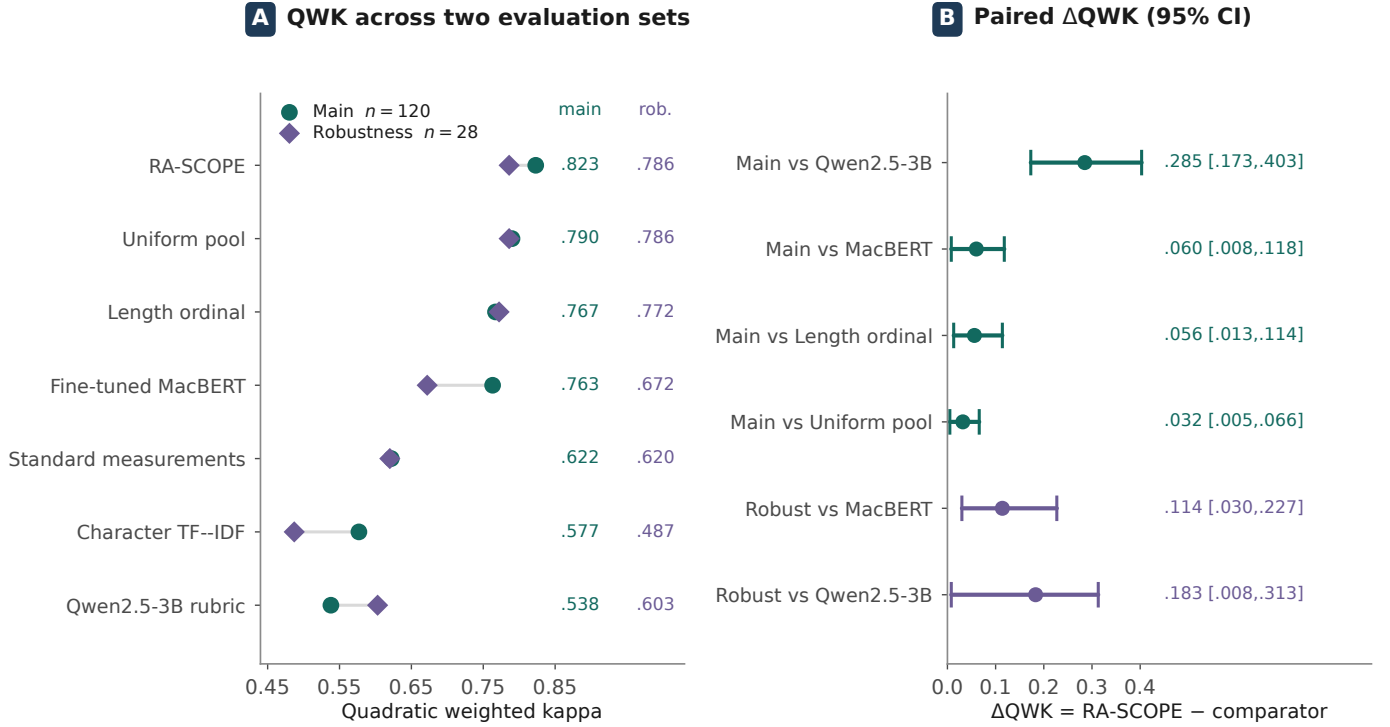


Fig. 2. Primary ordinal performance. (a) Main and robustness QWK across RA-SCOPE and six comparators; circles and diamonds distinguish the two independently adjudicated collections. (b) Direct paired QWK gains for the four prespecified main comparisons and two robustness comparisons; bars are 95% lineage-bootstrap intervals.

absolute error. Its QWK is .844 against .823, a difference of $+.021$ (95% book-bootstrap CI $[-.026, +.071]$, that is, not distinguishable at this sample size), while its MAE is .277 against .417, a difference of $-.140$ (CI $[-.248, -.035]$), and its catastrophic-error rate is .001 against .008. The advantage is stable across all four fold counts, with QWK between .844 and .848.

The comparison is not a like-for-like substitution, and the difference is precisely the property RA-SCOPE was designed for. The stacked model is fitted with access to target-collection labels: it cannot be constructed for a platform on which no adjudicated books exist, and its per-item decision rule is a fitted weight vector rather than an inspectable contract. RA-SCOPE consumes no target labels and every activated consensus grade remains provably preserved. We therefore read Table IV as the measured price of auditability on this collection rather than as a defect of the contract: on MAE the contract pays about .14 levels while retaining the guarantee, the state semantics and cold-start operation.

D. Expert Complementarity and Probability Quality

The ensemble is not averaging replicas. Its four experts disagree on 104/120 main books, with mean pairwise disagreement .576 and 42 distinct prediction patterns. Each expert is uniquely exact on at least four books (length 4, character 12, measurements 10, MacBERT 6), and at least one expert is exact on 85.8% of books. RA-SCOPE is exact on 60 disagreement cases, four more than uniform pooling, including

TABLE IV
LEARNED FUSION VERSUS THE ZERO-PARAMETER CONTRACT. THE STACKED ROW IS THE MEAN OVER TEN SEEDS OF A LINEAGE-GROUPED CROSS-VALIDATION AT $k=5$; THE DIFFERENCE ROW IS THE DIFFERENCE OF THE TWO ROWS ABOVE.

Method	QWK	MAE	Exact	> 1
RA-SCOPE (no target labels)	.823	.417	.592	.008
Stacked logistic (5-fold CV)	.844	.277	.724	.001
Difference	$+.021$	$-.140$	$+.132$	$-.007$

one book on which no individual point prediction is exact. The robustness set shows a similar mean pairwise disagreement of .589.

Point accuracy and probability quality separate cleanly. Projection improves the uniform pool’s Brier score (.085 to .083), NLL (1.443 to 1.411) and selective RC-AUC (.437 to .418), with a small ECE trade-off (.187 to .203). Length has the best Brier/NLL (.080/1.306) but substantially worse ECE (.274) and selective risk (.543), and Qwen has the weakest values on all four (.129/2.020/.269/.749). The QWK gains therefore do not come from globally sharpening probabilities without regard to ranking or coverage.

E. Length, Source, and Level Stress Tests

RA-SCOPE’s continuous score has Spearman $\rho = .901$ with the gold level. After residualizing both variables for log character count, log page count and source, partial

$\rho = .400$ [.258, .545], whereas the length expert gives $-.034$ [−.091, .090]; their paired partial-correlation difference is [.238, .573]. On 24 disjoint counter-length pairs, where character count orders two books opposite to their gold levels, RA-SCOPE orders the pair correctly with .708 accuracy [.500, .875], compared with .000 for length by construction. On 56 nearest different-level disjoint pairs it reaches .875 [.786, .946]. Source slices are consistent with the aggregate result: StoryWeaver ($n=89$) yields .775 QWK/.416 MAE and Global ASP ($n=23$) .795/.391; the eight-book GDL slice is descriptive (.543/.500). Band-wise MAE is .500 for L1–L2, .375 for L3 and .353 for L4–L5, below length and MacBERT in every band.

F. Construct and Modality Evidence

The mean of the ten adjudicated criterion grades correlates .935 [.907, .952] with the global grade. Controlling log characters, log pages and source leaves partial $\rho = .720$ [.589, .800], and leaving out any single criterion keeps the raw correlation between .920 and .941. This convergent evidence links the global target to the Standard’s operational dimensions rather than to length alone. For criterion prediction, observable-modality specialists improve macro-QWK from .456 to .532, a paired gain of .076 [.055, .096], and reduce macro-MAE by .119 [−.158, −.082]. DINOv2 illustration features [33] improve illustration-independence QWK from .309 to .365 and MAE from 1.092 to .925; deterministic pinyin features improve QWK from .260 to .966 and MAE from 1.125 to .100. Layout remains on the text route because source families expose incomparable page geometry.

VII. DEPLOYMENT AND HUMAN-IN-THE-LOOP REVIEW

RA-SCOPE is designed to sit inside an existing review process rather than to replace it, and the measurements above translate directly into operating behavior.

Cost. The layer itself is closed-form and adds negligible overhead to the four experts it composes. The expensive components are the experts: MacBERT training used one RTX 4060 Laptop GPU for approximately 7.7 GPU-minutes across three seeds, and the Qwen2.5-3B rubric baseline scored 244 books in 237 seconds using bfloat16 with 8.93 GB peak allocated GPU memory. The pooled projection is a coordinate-wise clip (Eq. 7) and can be re-executed by a reviewer by hand.

Triage. Accuracy stratifies by state. On the main collection, CONSENSUS items are exact 71.1% of the time and are never more than one level away ($n=38$, within-one accuracy 1.000), whereas CONFLICT and NO-SUPPORT items are exact 56.5% and 50.0% of the time ($n=46$ and $n=36$). A workflow can therefore accept CONSENSUS grades under a one-level tolerance and route CONFLICT and NO-SUPPORT items to human adjudication, which concentrates scarce expert time on 82 of 120 books. Nothing is hidden: every item still receives an ordinal distribution, and the state is reported alongside it.

Preservation. The guarantee in Proposition 2 is what makes this routing defensible. When the two anchors agree and at least one is support-eligible, the projection provably returns that agreed grade: observed as 38/38, 32/32 and 7/7 preservation on the main, source out-of-fold and robustness collections. Adopting a CONSENSUS grade therefore cannot overturn a decision that the evidence already supported.

Cold start on a new platform. Fitting the experts and calibrating the support rank requires only the 96-book source-development collection; no labels from the new platform are needed. This is where the zero-parameter design pays for itself, because the stacked fusion of Section VI-C cannot be constructed at all until adjudicated books exist on the target platform. Source-slice results (.775 QWK for StoryWeaver, .795 for Global ASP) indicate that the same configuration transfers across platforms, and the per-item support state makes any residual misfit visible instead of silent.

VIII. DISCUSSION

RA-SCOPE separates three questions that ordinary fusion merges: what the experts propose, whether anchor evidence resembles the source, and which decisions the ensemble is allowed to alter. The support rank is label-free; full agreement prevents a singleton from masquerading as consensus; and the convex projection turns the decision rule into a unique, executable operation. These properties make the guarantee inspectable per book while retaining a complete ordinal distribution.

The empirical pattern is equally specific. Uniform pooling already supplies a strong heterogeneous baseline. Projection changes only nine main-set absolute errors relative to that pool (seven improve, two worsen), and this is enough for a positive paired QWK interval and halves the frequency of errors above one level. Gains over length survive covariate control and counter-length comparisons, while the Qwen rubric baseline shows substantial platform sensitivity despite explicit instructions not to use length alone. Capability routing further shows that illustration and pinyin criteria benefit from evidence unavailable to a text-only global model.

Why zero-parameter pooling rather than learned fusion. Zero-parameter pooling is a design requirement, not a convenience. Source platforms differ sharply in layout, length and illustration conventions. A learned fusion layer must be fitted on labeled books, so it carries its fitting-platform label distribution into the next platform, and its per-item weights are not inspectable. RA-SCOPE’s fusion layer is itself unfitted: given the four expert chains, Eq. 7 can be re-executed and audited item by item, which a fitted weight vector does not permit. Where a fitted alternative is affordable it does raise average fit: the stacked fusion of Section VI-C reduces MAE by .140 relative to the contract. Our claim concerns decision guarantees and auditability under platform shift, not the best average fit on a labeled target collection.

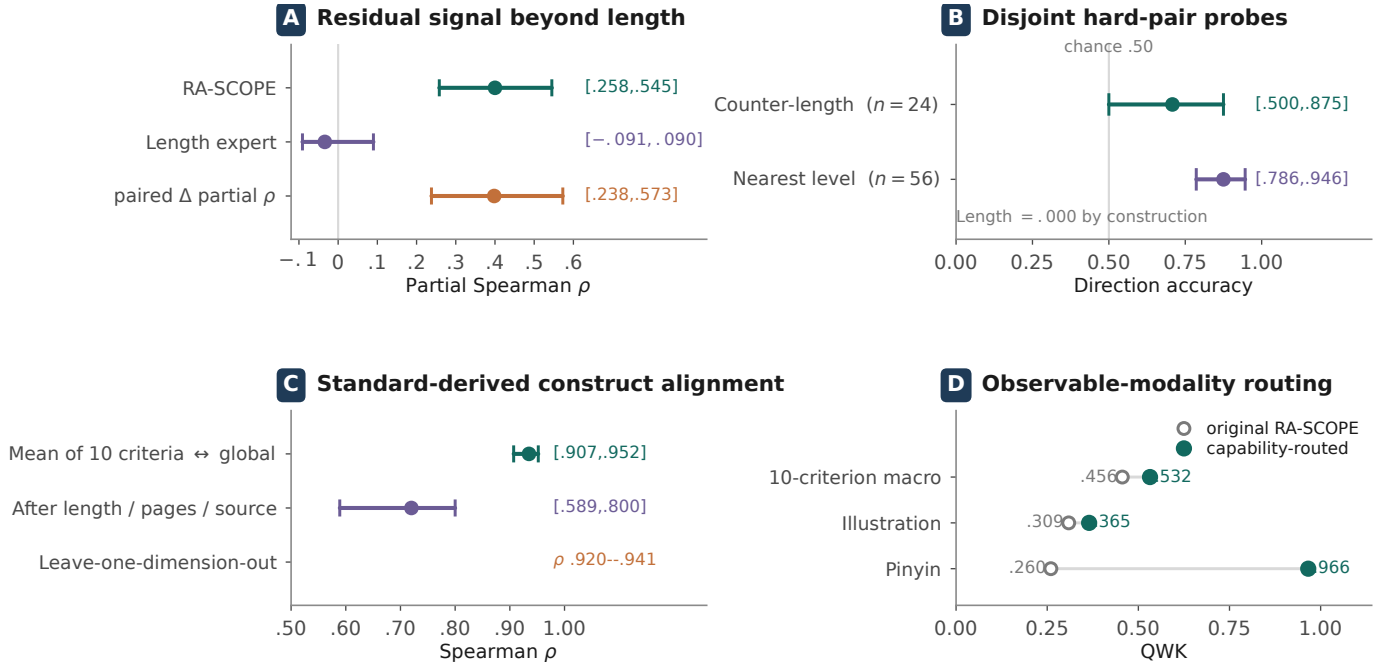


Fig. 3. Validity beyond aggregate accuracy. (a) Partial Spearman correlations after controlling length, page count and source. (b) Direction accuracy on disjoint counter-length and nearest different-level pairs. (c) Convergent alignment between the global grade and Standard-derived criteria. (d) Criterion-level gains from routing illustration and pinyin to observable-modality specialists.

IX. LIMITATIONS AND ETHICAL CONSIDERATIONS

Observed gold labels contain four L5 books and no L6 books, and the 28-book robustness collection is small; the results therefore support L1–L4 use most directly, with L5 descriptive and L6 untested. Material grades are not learner-comprehension outcomes. Architecture selection had access to the adjudicated collections, so a prospectively untouched, upper-level sample remains necessary for deployment-level validation. The conformal interpretation of the support rank requires exchangeable calibration and target representations, and platform shift can violate that assumption; a high support rank does not imply prediction correctness, and two supported anchors can share an error. StoryWeaver dominates the main collection, while the smallest source slice has eight books. The Qwen experiment is a reproducible 3B zero-shot rubric baseline, not a claim about all larger or proprietary LLMs. OCR, page extraction and DINOv2 style sensitivity can introduce source-specific error, and comparable geometric evidence is still needed for a dedicated layout specialist.

The study uses books and adult professional annotations, with no learner records or children’s personal data. Annotators are represented by role identifiers, and independent ratings precede separate adjudication. Predictions are intended to assist human material review: conflict, no-support and upper-band cases remain visible rather than being presented as certified decisions. Restricted book text and illustrations remain with their source providers, and users must preserve applicable attribution and license conditions. The Standard may privilege standard written Mandarin and conventional print layouts;

dialectal, culturally specific, accessibility-relevant and visually driven materials require human validation.

X. CONCLUSION

We presented a support-backed full-consensus projection for ordinal picturebook grading. It combines four monotone experts without fitted fusion parameters, attaches an explicit support state to every item, and uniquely preserves every activated anchor-consensus grade. Across source-only out-of-fold, main and separate robustness evaluations it improves or matches the strongest individual experts and the uniform pool while remaining stable across support and margin settings. The result is a compact inference architecture whose accuracy, decision semantics and evidence boundary can all be audited, and whose states translate directly into human review triage.

ACKNOWLEDGMENT

This work was supported by the National College Students’ Innovation and Entrepreneurship Training Program (No. 202610032033). The reproducibility artifact (the projection module, its invariant tests, every per-book prediction, the reported tables and the figure sources) is available at <https://github.com/yyting2006-ai/RA-SCOPE-artifact>.

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